

Review

Artificial intelligence in thermodynamics: hybrid modeling of thermophysical properties of fluids

 Hans Hasse, Sebastian Schmitt and Fabian Jirasek^{*}


Artificial intelligence is currently transforming thermodynamics. Hybrid models that combine machine learning (ML) with physical modeling enable predictions of thermophysical properties with unprecedented scope and accuracy. Focusing on the thermophysical properties of fluids, recent advances in this field are highlighted, covering two hybridization techniques: (i) embedding ML into physical models and (ii) incorporating physical knowledge into ML models. The discussion covers different types of thermodynamic models: (i) excess Gibbs energy models, (ii) equations of state, and (iii) force field models. The new hybrid models combine the soundness of physical models with the flexibility of ML models and give the best results when trained on large data sets, which are, however, not always available. The new hybrid models often significantly outperform widely used classical physical thermodynamic benchmark models. We have only begun to explore the new routes opened up by hybrid thermodynamic modeling; this review provides a starting point for future work in this field.

Address

Laboratory of Engineering Thermodynamics (LTD), RPTU Kaiserslautern, Kaiserslautern, Germany

Corresponding author: Jirasek, Fabian (fabian.jirasek@rptu.de)

^{*} Laboratory of Engineering Thermodynamics (LTD), RPTU Kaiserslautern Erwin-Schrödinger-Straße 44, 67663 Kaiserslautern, Germany

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Introduction

Thermophysical properties are foundational in chemical engineering; they determine which processes are feasible and which are not, and they are decisive for the economics and sustainability of processes. However, experimental data are notoriously scarce, so accurate and reliable

prediction methods of thermophysical properties are essential. We focus here on the thermophysical properties of fluids, with special attention to liquid mixtures, which are pivotal in chemical engineering practice.

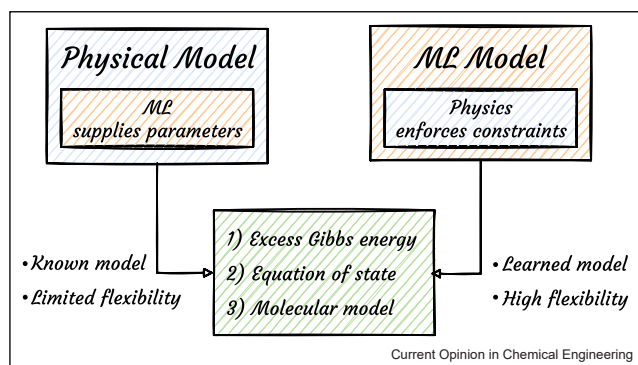
In recent years, data-driven approaches using artificial intelligence (AI) and machine learning (ML) have emerged as a powerful alternative to physical models for predicting thermophysical properties. Neural networks (NN) [1,2] and other ML models, such as matrix completion methods (MCM) [3–6], can learn complex patterns directly from data and have achieved strong prediction accuracy for many properties.

Three key factors influence the performance of ML models for thermophysical property prediction: (a) the available training data, (b) the molecular descriptors used as component representations, and (c) the ML model architecture. While different types of molecular descriptors, ranging from molecular fingerprints [7], text strings, like SMILES [8] and SELFIES [9], to molecular graphs, as well as different ML model architectures, like transformers [10] or graph neural networks (GNNs) [11], have been used (see [12,13] for in-depth discussions), the relatively small experimental data sets for thermophysical properties available for model training remain a critical issue, significantly impairing the performance of data-driven ML models. Specifically, thermophysical data sets usually comprise less than 1 million data points, which is several orders of magnitude smaller than used for the training of LLMs, which can contain trillions of tokens [14]. Furthermore, purely data-driven models risk violating physical constraints and can behave unreliably outside their training domain [15,16], which significantly undermines their trustworthiness and acceptance, ultimately preventing their practical application.

Consequently, the emerging consensus is that *hybrid models* combining ML with physical knowledge represent the most promising path for thermophysical property prediction, enabling the fusion of the flexibility of ML with thermodynamic rigor [15,17–19]. In this paper, we provide a concise review of recent advances in hybrid thermodynamic model development, focusing on foundational ideas that enhance prediction accuracy and scope while maintaining thermodynamic consistency.

We discuss the impact of ML on different established classes of thermodynamic models, namely, excess Gibbs

Figure 1



Different strategies for developing hybrid thermodynamic models and applications covered in this article.

energy models, equations of state, and force fields for molecular simulations, focusing on the two most important types of hybrid thermodynamic models: i) Models that embed ML methods in a physical modeling framework, and ii) ML models into which physical knowledge is infused explicitly, cf. Figure 1. For each class, we highlight important recent developments, primarily from the last 2–3 years, and discuss how these approaches achieve accuracy, applicability, and consistency.

Hybrid excess Gibbs energy models

Models of the excess Gibbs energy g^E are workhorses for describing the non-ideality of liquid mixtures and are firmly established in both academia and industry for modeling phase and reaction equilibria. The most widely used g^E -models, such as NRTL [20], UNIQUAC [21], and UNIFAC [22], are lattice models. Also, the models of the thermodynamic COSMO family are used, in which the energetic term of the excess Gibbs energy is calculated from quantum-chemical information [23]. The activity coefficients derived from g^E -models satisfy exact thermodynamic relations, such as the Gibbs-Duhem equations, by construction.

Embedding machine learning in physical models of the excess Gibbs energy

One strategy to enhance conventional g^E models is to train an ML model to predict the g^E model's pair-interaction parameters. The first hybrid model of this type was introduced in 2022 [18] by embedding a MCM [3] into the framework of UNIQUAC [21], and training it to predict fundamental pair-interaction energies between more than 1100 molecular components. MCMs are a family of ML methods to predict the unknown entries of a partially filled matrix, for example, pair-interaction parameters, by learning and exploiting similarities between the instances that span the matrix, for example, components, based on the available data. The hybrid

approach extended the applicability of UNIQUAC by orders of magnitude and outperformed the most powerful public version of UNIFAC (modified UNIFAC (Dortmund)) [24,25] in the accuracy of the predicted activity coefficients. The downside of this hybrid model lies in the nature of the MCM, which can only generalize inside the matrix spanned by the components from the training set and, hence, does not allow extrapolations to mixtures containing completely unstudied components.

This limitation has been addressed by using transformer models, an architecture that significantly advanced ML by enabling the simultaneous processing of entire sequences of text to understand context. Winter et al. [26] trained a transformer model (SMILES-to-Properties Transformer (SPT)) to output parameters of the NRTL model [20] for basically any pair of molecules, as only their structural formula in the form of SMILES is required as input. Their SPT-NRTL was trained not only on experimental data but also on an extensive set of synthetic activity coefficient data obtained from COSMO-RS [23] and achieves higher prediction accuracy than modified UNIFAC (Dortmund) [25] for a wide range of mixtures. While the authors did not publish their full model, they provide a set of NRTL parameters for 100 million pairs of molecules, greatly extending the application range of NRTL. Similarly, Sanchez-Medina and Sundmacher [27] successfully demonstrated the suitability of GNNs for predicting parameters of the Margules g^E model.

Recently, Hayer et al. [28,29] transferred the embedding approach to different versions of the UNIFAC group-contribution model using MCMs for completing pair-interaction parameter sets by holistic training on an extensive set of experimental activity coefficient and excess enthalpy data. The new parameterizations, labeled UNIFAC 2.0 and modified UNIFAC 2.0, not only significantly extend the scope of UNIFAC but also substantially increase its prediction accuracy. They were fully published and have been adopted by major providers of process simulation software, such as AspenTech and AVEVA. They were also included in the software package of the Dortmund Data Bank.

Physics-informed machine learning models of the excess Gibbs energy

A promising alternative to embedding ML methods into established physical models is the inverse approach, which incorporates explicit physical knowledge into flexible ML algorithms. Different ways to introduce physics into AI models have been proposed [16,15].

One very successful approach is the physics-informed neural network (PINN) framework [30]. In PINNs, the loss function includes, besides a classical loss penalizing deviations from the training data, terms that penalize

deviations from known constraints. A notable example of PINNs in thermodynamics is provided by Rittig et al. [31], who developed a GNN for predicting concentration-dependent activity coefficients in binary mixtures that was penalized for deviations from the Gibbs-Duhem equation during training. The authors demonstrated that the PINN approach significantly reduced physical inconsistencies compared to an ablation model, while also increasing prediction accuracy.

However, soft constraints, as in the PINN approach, only yield *approximate compliance* with physical laws, which might be unacceptable in practice, and they always entail a trade-off between consistency and accuracy. Consequently, *hard-constraint* models, where the model structure is designed such that it inherently satisfies the relevant physical laws, with no freedom to violate them, are preferred. Rittig and Mitsos [32] and Specht et al. [33] developed hard-constraint NNs for predicting activity coefficients in binary mixtures. Rittig et al. [32] used a GNN to generate molecular embeddings and trained their model on synthetic activity coefficient data from COSMO-RS [23] and enforced only Gibbs-Duhem consistency. By contrast, Specht et al. [33] relied on the transformer ChemBERTa-2 [34] to generate molecular embeddings from SMILES and trained their model, HANNA, on the extensive set of experimental activity coefficients from the Dortmund Data Bank and enforced compliance with the entire set of relevant physical laws. HANNA needs only SMILES as input, giving it an unprecedented scope. It also outperforms modified UNIFAC (Dortmund) [25] in prediction accuracy. Very recently, HANNA has been extended to direct training on liquid-liquid phase equilibrium data and extrapolations to multi-component mixtures [35], enabling, for the first time, the accurate and holistic predictive modeling of vapor-liquid, liquid-liquid, and vapor-liquid-liquid equilibria in basically any mixture of molecular components. However, the most important practical application of models of excess Gibbs energy remains the calculation of vapor-liquid equilibria, which also requires information on the pure-component vapor pressures. Recently, Hoffmann et al. [36] developed a hybrid method for predicting pure-component vapor pressures based solely on molecular graphs, which complements HANNA perfectly.

Hybrid equations of state

Modern equations of state (EoS) are fundamental equations and enable the calculation of all thermodynamic properties of pure components and mixtures in the entire fluid region. They are typically presented in the form of the residual Helmholtz energy, so that, additionally, information on the ideal gas state is needed, which can, however, be obtained from a priori quantum-chemical methods, even if no experimental data are

available. Furthermore, EoS can be extended to describe interfacial properties and transport properties [37,38]. There are EoS with a moderate number of parameters that can be chosen to describe a wide range of fluids and multi-parameter EoS that are designed for describing only a specific fluid [39]. Widely used EoS from the former type are those from the SAFT family, for example, [40–42], as well as the virial EoS (only for moderate densities) and cubic EoS, which continue to be used to date [43]. The parameters of all these EoS are usually fitted to experimental data of pure components. Combination and mixing rules, typically including additional adjustable parameters, are used to describe mixtures. In this section, we discuss how ML is integrated into EoS modeling to improve predictions of pure-component and mixture properties.

Embedding machine learning in physical equations of state

Most studies in this field aim at parameterizing existing EoS for pure fluids using NNs. Recent examples include the works of Winter et al. [44], Felton et al. [45], Habicht et al. [46], and Matsukawa et al. [47], who all worked with the PC-SAFT EoS or an extension, the PCP-SAFT EoS. Among these, the work of Winter et al. [44] stands out. In their 2025 study, they developed SPT-PC-SAFT. This transformer-based model infers the pure-component PC-SAFT parameters (segment number, segment diameter, dispersion energy, etc.) directly from a component's SMILES. Their resulting hybrid model was trained end-to-end using experimental vapor pressure and liquid density data for thousands of compounds, with PC-SAFT calculations embedded as a differentiable layer during the training. The model achieves high accuracy across a wide range of temperatures and pressures. The authors compared their model to group-contribution versions of PC-SAFT and found significant improvements. They did not disclose their model, but they released a database of predicted PC-SAFT parameters for over 13,000 molecules.

In contrast, Felton et al. [45], Habicht et al. [46], and Matsukawa et al. [47] utilized databases of PC(P)-SAFT parameters to develop their models, which cover only a few hundred components. This training on parameter sets is much simpler than the end-to-end training of Winter et al. [44], as it disentangles the calculation of the thermodynamic properties and the training of the ML method, but has the disadvantage that any flaws of the EoS parameterization will be carried over into the ML training. In general, one can expect better results from end-to-end training, as seen in Hayer et al. [28]. However, the integration of thermodynamic calculations into ML model development poses a significant challenge: standard routines, such as those for phase equilibrium calculations, cannot be used in most cases, necessitating the development of adapted solutions. An alternative is

to use surrogate models, which, in turn, can also be developed using ML, see, for example, Sun et al. (2024) [48].

Cretu and Perez-Rios [49] have developed a Gaussian process regression model for predicting second virial coefficients from molecular descriptors based on a dataset comprising data for about 9000 pure substances. Matsukawa and Otake [50] describe a NN to predict binary interaction parameters of a cubic EoS from simple descriptors; however, only for a limited class of mixtures.

Machine learning of new equations of state

ML can also support learning the EoS itself. Using pseudo-data generated from the SAFT-VR Mie EoS, Zhu and Müller [51] have demonstrated that ML techniques (Gaussian processes and NNs) can be used for describing critical properties, vapor pressures, and densities of a pure fluid without pre-selecting any functional form. Chaparro and Müller [52] extended this idea to develop a NN for predicting the residual Helmholtz energy of Mie fluids based on molecular simulation data, while incorporating physical constraints into the loss function. The accuracy of the approach matches that of the best analytical EoS for Mie fluids, while the flexibility of the NN enables even the description of solid states.

In an early work, Scalabrin et al. [53] have incorporated a NN into a simple Helmholtz energy EoS; however, without attempting to generalize over substances. Rosenberger et al. [54] demonstrate how automatic differentiation can be used for end-to-end training of Helmholtz energy EoS on various types of thermodynamic data and show its superiority compared to classical multitask predictions. Recently, Dijkman et al. [55] demonstrated that a NN trained solely on data for radial distribution functions can predict Helmholtz free energy density functionals that can be used for predicting interfacial properties. Furthermore, Yang et al. [56] have used symbolic regression to determine temperature-dependent functions for the various terms in a cubic EoS based on NIST's Refprop database containing reference EoS for about 150 pure components.

Machine learning in molecular simulations

Molecular simulations, employing techniques such as molecular dynamics and Monte Carlo, are a source of *in-silico* data for thermophysical properties. They can also be used to predict these properties in cases where experiments are infeasible. However, molecular simulations are computationally expensive, and the quality of the results depends on the availability of high-quality force fields, which limits the applicability of the approach. ML techniques are being used in two main ways to enhance molecular simulations: (i) to develop force

fields based on quantum-chemical calculations, (ii) to improve the efficiency of the molecular simulation runs. We focus here only on (i) and refer the reader to the reviews of Mehdi et al. [57] and Zhu et al. [58] regarding (ii).

Classical force fields are given as analytical functions that contain adjustable parameters fitted to experimental or quantum-chemical data. Recently, ML-based force fields (MLFFs) have emerged as an alternative. They are used in a wide range of applications, for example, as surrogate models to replace costly quantum-chemical calculations [59]. Here, we are interested solely in the application of MLFFs in molecular simulations for the determination of thermophysical properties, which is not the primary focus of most MLFF papers. In this context, an interesting application of MLFFs in thermodynamics is the prediction of properties of water [60,61], a task with which classical force fields struggle. Many authors have also developed MLFFs for predicting thermophysical properties of electrolyte solutions for battery technology [62]. In contrast to training MLFFs on quantum-chemical calculation data (bottom-up approach), using experimental (macroscopical) data for training MLFFs remains challenging. Differentiable and reversible molecular simulations provide a promising approach to overcome this challenge while being computationally efficient, as demonstrated by Greener [63].

Conclusions

As Venkatasubramanian [17] mused, the promise of AI in chemical engineering will be realized when it's no longer AI or first-principles, but AI and first-principles together. We fully agree and argue that thermodynamics provides an ideal proving ground for this synergy: it has a vast body of explicit physical knowledge, relatively large data sets, and, last but not least, direct relevance to many fields in science and engineering. The convergence of AI and thermodynamics is leading to powerful hybrid models that transcend the limitations of either approach alone. Different routes for combining both worlds, each with its advantages and challenges, await exploration.

From a broader perspective, integrating ML with physical knowledge not only improves performance but also provides *interpretability*. A black-box NN that predicts some thermophysical property might be accurate in certain situations. However, a hybrid model that outputs, say, the parameters of an EoS is more tangible to chemical engineers. This hybrid approach keeps one foot in the traditional modeling world, conserving the strengths of the underlying physical models, including physical consistency, robust extrapolations to multi-component mixtures, and mitigating the 'black box' concern that often surrounds ML: An engineer can use ML-predicted EoS parameters for a novel molecule and trust that the

resulting calculations will obey expected rules. Moreover, using established frameworks increases the interpretability of the resulting models and facilitates assessing the reliability of the model, since strengths and weaknesses are usually well understood and can be taken into account when applying the model. Similarly, process simulators can incorporate these hybrid models with minimal modifications, often by simply exchanging parameter sets. These advantages make the hybrid “ML predicts thermodynamic model parameters” paradigm attractive. However, this approach also comes with the price of adopting the limitations of physical models, such as limited flexibility in describing all types of mixture behavior.

Going further, the infusion of explicit physical knowledge into ML models is enabling a new generation of predictive thermodynamic models that take full advantage of the flexibility of ML but consistently and robustly extrapolate to unseen grounds. As these techniques mature, we expect most ML models for thermophysical properties to have built-in physical ‘guardrails’ ensuring fundamental laws are never violated. While this type of hybrid thermodynamic model holds the promise of revolutionizing our field, new challenges arise regarding the integration of these models into industrial workflows and process simulators, as their implementation and evaluation differ from those of classical models.

Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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